# H2 - Team T76 — Adil Mutali - 23.10.26

Problem: minimize $3y_1 + 5y_2 + 6y_3$

$(T_1)\ y_1 + y_2 \geq 1 \quad (T_2)\ y_2 + y_3 \geq 1 \quad (T_3)\ y_1 + y_3 \geq 1 \quad 0 \leq y \leq 1$

(a) Take $y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

- $(T_1)\ \frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓
- $(T_2)\ \frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓
- $(T_3)\ \frac{1}{2} + \frac{1}{2} = 1 \geq 1$ ✓
- $0 \leq \frac{1}{2} \leq 1$, the point is feasible
- Cost: $3 \cdot \frac{1}{2} + 5 \cdot \frac{1}{2} + 6 \cdot \frac{1}{2} = 1.500 + 2.500 + 3.000$
  $= 7.000$
- Lower bounds $1 \cdot T_1 + 4 \cdot T_2 + 2 \cdot T_3$ gives
  $(y_1 + y_2) + 4(y_2 + y_3) + 2(y_1 + y_3) = 3y_1 + 5y_2 + 6y_3$
  $\geq 1 + 4 + 2 = 7$
- Every feasible point costs at least 7;
  so $z_LP = 7.000$.

(b) - Round up: $(1, 1, 1)$. Feasible. Cost $3 + 5 + 6 = 14.000$
- Round down: $(0, 0, 0)$. $(T_1)$: $0 \geq 1$ is false, so it is infeasible
- Rounding each coordinate separately:

$(1, 1, 0) = 8.000 \quad (1, 0, 1) = 9.000 \quad (0, 1, 1) = 11.000 \quad (1, 1, 1) = 14.000$

(c) All 8 binary points ($y_1, y_2, y_3 \to$ cost):

| 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
|---|---|---|---|---|---|---|---|
| ✗ | 16 ✗ | 5 ✗ | 11 ✓ | 3 ✗ | 9 ✓ | 8 ✓ | 14 ✓ |

- Feasible points: 011, 101, 110, 111, so $z^* = 8.000$ at $(1,1,0)$.
- LP gap: $(z^* - z_{LP}) / z^* = (8-7)/8 = 0.125$
- Every feasible integer point ($12.5\%$) costs at least 8, while the LP optimum is $7 < 8$. Hence no integral point can be an optimal solution of the LP.

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